Parduman Gupta

+91-8514866197 | parduman909154@gmail.com | parduman123

Career Objective

- M.Tech (ECE) student with strong fundamentals in digital design and VLSI, and hands-on experience in RTL development of FIFO, FSM, and AXI-based interfaces during AMD internship and academic projects. Seeking a VLSI Engineering role to contribute to synthesizable hardware design, timing-aware implementation, and IP development in semiconductor systems.

Education

•Motilal Nehru National Institute of Technology, Prayagraj *2024-2026*
MTech-ECE CGPA: 9.55/10

•AKTU,Lucknow(REC, Bijnor) *2020-2024*
BTech-EE Percentage: 80%

•CRGCBI,Howrah *2018-2019*
WBCHSE, WestBengal Percentage: 88%

•CRGCBI,Howrah *2016-2017*
WBBSE, WestBengal Percentage: 89.57%

Experience

•Advanced Micro Devices(AMD) *Aug 2026 – Present*
RTL Design & Static Verification Engineer (Contarctual Role) Hyderabad

- Performed comprehensive **RTL static checks** including **CDC, RDC, Lint, STA, Synthesis, IDCLINT, DFT & Power, and LEC** checks on **all TSMC technology nodes** (N3, N4, N5, N7) designs.
- Worked on RTL quality analysis and improvement using AMD's **RQual framework and TileBuilder platform**, ensuring clean and robust RTL across multiple design blocks.
- Conducted **RTL violation debug, waiver management, and sign-off** activities to deliver synthesis-ready, high-quality RTL for advanced node designs.
- **Skills:** Conducted RTL violation debug, waiver management, and sign-off activities to deliver synthesis-ready, high-quality RTL for advanced node designs.

•Advanced Micro Devices(AMD) *July 2025 – June 2026(12 months)*
Intern Hyderabad

- Designed and developed **RTL** for **VCU2** and **ISP IP** using Verilog, implementing core functionality along with custom logic based on customer requirements
- Implemented **CDC (Clock Domain Crossing)** and **RDC (Reset Domain Crossing)** safe design techniques to ensure reliable multi-clock and reset domain operation
- Designed and optimized **FIFO architectures** for efficient data buffering, throughput handling and low latency management across modules
- Developed supporting **RTL blocks** and integrated multiple IP components to achieve complete system-level and IP-level functional requirements
- Performed **RTL validation** using Vivado (synthesis, implementation, and timing analysis), resolving timing violations and ensuring design closure
- Executed hardware validation on **VEK385 FPGA** platform, performing RTL-to-hardware correlation and verifying functionality on real silicon
- **Skills:** Verilog, RTL Design & Coding, FSM Design, CDC & RDC Synchronization, FIFO Design, Micro-architecture Design, Clock & Reset Architecture, Linting, Synthesis (Xilinx Vivado), Static Timing Analysis,Timing Closure Basics, FPGA Prototyping & Hardware Validation, Vivado (IP Integrator, Synthesis, Implementation),FPGA Hardware Validation

Personal Projects

•Synchronous & Asynchronous FIFO Design

*RTL Design of **Synchronous** and **Asynchronous FIFO** using **Verilog HDL**.*

- Designed and implemented Synchronous and Asynchronous FIFO modules in Verilog HDL using Xilinx Vivado, ensuring robust data buffering across single and multi-clock domains.
- Implemented CDC-safe Async FIFO architecture using Gray-code pointer encoding and dual flip-flop synchronizers to eliminate metastability.
- Conducted Clock Domain Crossing (CDC) analysis to identify and resolve multi-clock domain violations, ensuring reliable cross-domain data transfer.
- **Tools & Skills**: Verilog HDL, Xilinx Vivado, FIFO Architecture, CDC (Gray Code, 2-FF Synchronizers), STA, RTL Design

•FSM Sequence Generator and Detector Design

Mealy FSM Non-Overlapping Sequence Detector using ***Verilog HDL***

- Designed a Mealy-based finite state machine to detect a non-overlapping input sequence using Verilog HDL.
- Defined state transitions and output logic to ensure correct detection and reset behavior.
- Validated functionality with self-written testbenches and protocol-compliant transactions.
- **Tools & Skills:** Verilog HDL, FSM Design (Mealy), RTL Simulation, EDA Playground.

•STA of 2-Bit Synchronous Counter

Static Timing Analysis of a 2-Bit Synchronous Counter using ***Synopsys PrimeTime***

- Performed Static Timing Analysis (STA) of a 2-bit synchronous counter using Synopsys PrimeTime, applying SDC constraints (create_clock, I/O delays, clock uncertainty).
- Analyzed setup/hold timing, identified critical paths, and evaluated slack for timing closure.
- Studied clock latency, skew, and uncertainty effects, and debugged timing violations through iterative STA runs with path-based analysis.
- **Tools & Skills:** Verilog, Synopsys PrimeTime, SDC & Basic TCL commands.

Technical Skills and Interests

Core Skills	RTL Design | Digital Design | FIFO & FSM Architecture | STA (Setup, Hold, OCV, CRPR) | CDC (Gray Code, 2-FF Synchronizers) | ASIC Design Flow | RTL-to-GDSII Understanding | Low-Power Design Techniques & UPF (Unified Power Format) | SPEF Analysis | SDC Constraints
HDL & Languages	Verilog HDL | Verification (Testbench Development, Debugging, Coverage Analysis) | TCL Fundamentals
EDA Tools	Xilinx Vivado | Synopsys PrimeTime | Synopsys VCS | Synopsys SpyGlass | Linux (Vim, Shell)
Protocols	AXI4-Stream | AXI4 (IP Integration)
Domain Knowledge	Digital IC Design | Low Power Design | Computer Architecture | SoC Architecture | Gate-Level Simulation
Soft Skills	Strong Problem-Solving Mindset | Attention to Detail | Effective Communication | Team Collaboration | Cross-functional Collaboration | Fast Learner
Areas of Interest	VLSI & ASIC Design | SoC Architecture | Semiconductor Technologies
Languages	English | Hindi | Bengali

Achievements

•Educational Achievement

- Qualifying **GATE EE 2024** in Final Year
- Ranked in the **top 5** of my college throughout my B.Tech.
- Achieved **8th Rank** (12th BOARD) in Uluberia.

•Extra-curricular Achievement

- Secured 2 Silver Medals at National-Level Karate Championships (2014, 2015).
- Active participant in cultural and spiritual programs at university level.